Community Diagnosis

Goal: Improve skills and knowledge on community diagnosis

Specific objectives:

- Define the concepts of community diagnosis and its purposes
- Describe the steps in carrying out community diagnosis
- Conduct a participatory rural appraisal as a tool for community diagnosis

---

Community – is a cluster of people with at least one common characteristic (geographic location, occupation, ethnicity, housing condition). It can also be defined as a group of people with a common characteristic or interest living together within a larger society

Diagnosis – is the identification and confirmation of a problem

Community sub systems - A community is a whole entity that functions because of the interdependence of its parts or subsystems. Community core are: history, socio-demographic characteristics, vital statistics, values/beliefs/religions. Community sub systems are: Physical environment; Education; Safety and transportation; Politics and government; Health and social services; Communication; Economics; and Recreation.

Community diagnosis

- Refers to the identification and quantification of health problems in a community as a whole in terms of mortality and morbidity rates and ratios, and identification of their correlates for the purpose of defining those at risk or those in need of health care.

- Is a comprehensive assessment of health status of the community in relation to its social, physical and biological environment
- According to WHO definition, it is "a quantitative and qualitative description of the health of citizens and the factors which influence their health. It identifies problems, proposes areas for improvement and stimulates action

Purpose - The purpose of community diagnosis is to define existing problems, determine available resources and set priorities for planning, implementing and evaluating health action, by and for the community. The following are the importance of community diagnosis:

- Analyze the health status of the community
- Evaluate the health resources, services, and systems of care within the community
- Assess attitudes toward community health services and issues
- Identify priorities, establish goals, and determine courses of action to improve the health status of the community
- Establish an epidemiologic baseline for measuring improvement over time.
- To act as a data reference
- To provide an overall picture of the local community and the residents' concerns
- To suggest priority areas for intervention and the feasible solutions
- To indicate the resource allocation and the direction of work plans
- It helps to find the common problems or diseases, which are troublesome to the people and are easily preventable in the community.
- It is a tool to disclose the hidden problems that are not visible to the community people but are being affected by them.
- It helps to access the group of underprivileged people who are unable to use the

available facilities due to poverty, prevailing discriminations or other reasons.

- It helps to find the real problems of the community people which might not have perceived by them as problems.

Implementation framework and process

Types of information in community diagnosis and intervention sources

- The information collected in a community diagnosis is called health indicators
- Characteristics of indicators:
  - **Valid,** i.e., they should actually measure what they are supposed to measure;
  - **Reliable** and objective, i.e., the answers should be the same if measured by different people in similar circumstances;
  - **Sensitive,** i.e., they should be sensitive to changes in the situation concerned,
  - **Specific,** i.e., they should reflect changes only in the situation concerned,
  - **Feasible,** i.e., they should have the ability to obtain data needed, and;
  - **Relevant,** i.e., they should contribute to the understanding of the phenomenon of interest.

Classifications of indicators:

- Mortality indicators; Morbidity indicators ; Disability rates ; Nutritional status indicators; Health care delivery indicators ; Utilization rates ; Indicators of social and mental health ; Environmental indicators; Socio-economic indicators ; Health policy indicators ; Indicators of quality of life' Other indicators

Mortality indicators are:	Morbidity indicators are:
o Crude death rates o Specific death rates: age/disease o Expectation of life o Infant mortality rate o Maternal mortality rate o Proportionate mortality ratio o Case Fatality rate	o Incidence and prevalence o Notification rates o Attendance rates: out-patient clinics or health centers. o Admission and discharge rates o Hospital stay duration rates
Disability indicators are:	Nutritional indicators are:

o No. of days of restricted activity o Bed disability days o Work/School loss days within a specified period. o Expectation of life free of disability	o Anthropometrics measurements o Height of children at school entry o Prevalence of low birth weight o Clinical surveys: Anaemia, Hypothyroidism, Night blindness

Health care delivery indicators are: o Doctor / Population ratio o Doctor / Nurse ratio o Population / Bed ratio o Population / per health center	Utilization indicators are: o Proportion of infants who are fully immunized in the 1st year of life i.e..immunization coverage. o Proportion of pregnant women who receive ANC. o Hospital-Beds occupancy rate. o Hospital-Beds turn-over ratio
Social/mental health indicators are: o Suicide & Homicide rates o Road traffic accidents o Alcohol and drug abuse	Environmental health indicators are: o Measures of Pollution o The proportion of people having access to safe water and sanitation facilities o Vectors density
Socio-economic indicators are: o Rate of population increase o Per capita income o Level of unemployment o Literacy rates - females o Family size o Housing condition e.g. No. of persons per room	Health policy indicators are: o Proportion of resources spent on health services. o Proportion of funds spent on health related activities. o Proportion of total health resources devoted to primary health care

Community diagnosis process/steps

Implementation of community diagnosis takes stages:

- Initiation stage
  - Set up a dedicated committee or working group to manage and coordinate the project.
  - The committee should involve relevant parties such as government departments, health professionals and non-governmental organizations.

- Identify the available budget and resources to determine the scope of the diagnosis.
- Some of the common areas (scope) to be studied may include health status, lifestyles, living conditions, socioeconomic conditions, physical and social infrastructure, inequalities, as well as public health services and policies.
- Once the scope is defined, a working schedule to conduct the community diagnosis, production and dissemination of report should be set.

- Data collection and analysis stage
  - Collect both quantitative and qualitative data.
  - Population census and statistical data (e.g. population size, sex and age structure, medical services, public health, social services, education, housing, public security and transportation, etc.) can provide background of the district.
  - Collect community data by conducting surveys through self-administered questionnaires, face to face interviews, focus groups and telephone interviews.
  - In order to ensure reliability of the findings, an experienced organization such as an academic institute can be employed for conducting the study.
  - The sampling method should be carefully designed and the sample size should be large enough to provide sufficient data to draw reliable conclusions.
  - Analyze and interpret the data.
  - Here are some practical tips on data analysis and presentation:
    - statistical information is best presented as rates or ratios for comparison
    - trends and projections are useful for monitoring changes over a time period for future planning
    - local district data can be compared with other districts or the whole population
    - graphical presentation is preferred for easy understanding

- Diagnosis stage
  - Diagnosis of the community is reached from conclusions drawn from the data analysis.
  - It should preferably comprise three areas:
    - Health status of the community
    - Determinants of health in the community
    - Potential for healthy community development

- Dissemination stage
  - The production of the community diagnosis report is not an end in itself
  - Efforts should be put into communication to ensure that targeted actions are taken.
  - The target audience for the community diagnosis includes policy-makers, health professionals and the general public in the community.
  - The report can be disseminated through the following channels:
    - ✓ Presentations at meetings of the health boards and committees, or forums organized, local community groups and the general public
    - ✓ Press release
    - ✓ Thematic events (such as health fairs and other health promotion programmes)

Evaluation of community diagnosis

- It is advisable to carry out evaluation after completing the community diagnosis.
- The following are the importance of evaluating community diagnosis process:
  - Identification of what diagnostic approaches that worked best versus those never worked

- It provides answers to the following questions: Did the diagnosis reveal the hidden problem in the community? Was the process interfered with by biasness? Were there missing responses in the questionnaires? Was the correct sample size used? What is the accuracy level of the responses? What can be improved in future diagnosis processes?

Principles of community based research

- Community based research is a research that strives to be:
  - ✓ *Community situated* -begins with a research topic of practical relevance to the community (as opposed to individual scholars) and is carried out in community settings.
  - ✓ *Collaborative* -community members and researchers equitably share control of the research agenda through active and reciprocal involvement in the research design, implementation and dissemination.
  - ✓ *Action-oriented* -the process and results are useful to community members in making positive social change and to promote social equity.
- The following are its principles:
  - ✓ Recognizes community as a unit of identity
  - ✓ Builds on strengths and resources within the community
  - ✓ Facilitates collaborative partnerships in all phases of the research
  - ✓ Integrates knowledge and action for the mutual benefit of all partners
  - ✓ Promotes a co-learning and empowering process that attends to social inequalities and different perspectives
  - ✓ Disseminates findings and knowledge gains to all partners in an accessible way.
  - ✓ Involves long term commitment by all partners
  - ✓ The relevance of the research topic is identified or verified by community members.
  - ✓ The resources of research (financial, expertise, etc.) are shared with community members, particularly those most affected by the research topic.
  - ✓ The research process recognizes and addresses power imbalances between researchers and community members.
  - ✓ The research process is driven by values, including: empowerment, supportive relationships, social change, learning as an ongoing process and respect for diversity.
  - ✓ The research process and results are accessible and understandable to community members.
  - ✓ The research process and results consider and adapt to the context in which the research is conducted.
  - ✓ The research leaves a legacy, both in terms of the utilization of research results, as well as in the future collaboration among partners.

Roles of households and communities

- What are the roles of the following in community diagnosis? **{Give as group work}**
  - Individual ; Family ; Institutions; Faith based organization; Leaders

Participatory rural appraisal

- Is a technique typically used in the field to gather qualitative data, often to complement quantitative data
- Its **aims** are:

- ✓ To involve local people in the selection, design, planning and implementation of programmes and projects that will affect them
- ✓ Ensures that local perception, attitudes, values and knowledge are taken into account as fully as possible.
- ✓ To make continuous and comprehensive feedback an integral part of all development activities.
- ✓ It acts as alternative to the two common qualitative methods:
  - o Questionnaires which often proved lengthy, costly and prone to errors,
  - o Rushed site visits by researchers to collect haphazard data from local elites.
  - o PRA uses a combination of approaches and methods to enable rural people to share enhance and analyze their knowledge of life and conditions, to plan and to act.
- ✓ Uses a range of methods to ensure reliability and validity, and to enable cross-checking
- ✓ Facilitate rapid and progressive learning with and from rural people
- ✓ Enables vulnerable groups in a community to have a voice and impart their views on issues of transportation and access from which they are most often excluded.
- ✓ Include: women, the elderly, disabled and even school children, researchers and other professionals are able to paint a realistic picture of community life; and through use of the different PRA techniques,
- ✓ It answers the following questions relevant to community needs:
  - o Which community members require the use of transport?
  - o When is access required?
  - o Why is access required?
  - o Where to local people travel to?
  - o How is transportation paid for?
  - o What modes of transport to people take?

- PRA methods are based on the following **principles**:
  - ✓ A reversal of learning - to learn with and from rural people, directly, on the site, face to face, gaining from local, physical, technical and social knowledge.
  - ✓ Learning rapidly and progressively - with flexible use of methods, improvisation, iteration, and cross-checking, being adaptable in a learning process.
  - ✓ Seeking diversity - Looking for, noticing and investigating contradictions, anomalies and difference.
  - ✓ Triangulating – use of combined approaches for data collection
  - ✓ Facilitating by the local people - Facilitating, investigation, analysis, presentation and learning by rural people themselves, so that they present and own their own outcomes.
  - ✓ Preparation - Ensure that all available secondary data on the locality and subject has been reviewed, allowing suitable villages to be identified to capture a broad sample, before surveying commences.
  - ✓ Facilitation - It is imperative that the external professional displays good facilitation skills, which aims to enable local people to undertake some or all of the investigation, mapping, modeling, diagramming, ranking, scoring, quantification, analysis, presentation and planning themselves.
  - ✓ Behavior and attitudes - The behavior and attitudes of external facilitators are of primary importance, more important than methods even. All important attitudes include: critical self-awareness and embracing error, sitting down, listening and

learning, not lecturing but allowing the villagers to be the main teachers and analysts.

- ✓ Longevity - Participatory approaches are not substitutes for, but are rather an integral part of, long term dialogue and sustained interaction. A single, brief participatory exercise with a group of local people will not lead to positive and lasting change.

- The following are **methods** of PRA
  - Triangulation
    - ✓ Triangulation is simply a method of linking different survey methods in order to crosscheck the information collected from each method.
    - ✓ Triangulation is of particular use when employing mapping, ranking and scoring, flow diagrams, Venn diagrams and wealth ranking techniques, as these involve group participation which may require some verification.

  - Observation
    - ✓ Prior to conducting any PRA techniques, researchers should be clear in their mind as to what exactly it is they are researching, and to have some realistic objectives of the PRA surveys.
    - ✓ Researchers should act on what they see, and recognize distinctions in gender, age, and wealth etc. amongst the community observation also aids improvisation, particularly when carrying out diagramming techniques, allowing household implements for example to be used in the PRA methods.

  - Semi-structured interviews
    - ✓ Participatory methods have contributed to adjusting the interview to make it more conversational, while still controlled and structured.
    - ✓ This is whereby only some of the questions and topics are predetermined, whilst the majority of questions will be formulated during the interview.
    - ✓ Questions are asked according to a flexible checklist and not from a formal questionnaire.

  - Ranking and scoring techniques
    - ✓ Ranking and scoring have long been used to assess people's expectations, beliefs, attitudes, preferences and opinions.
    - ✓ Ranking and scoring means placing something in order:
      - Ranking: putting in order
      - Scoring: weighting differences

  - Matrix ranking
    - ✓ Matrices can be used to express other issues, where a two dimensional comparison is useful, such as where frequency or occurrence can be illustrated in the same way as preference scoring.
    - ✓ Matrix ranking involves listing the elements down one side, and the criteria on which they are judged, gained from informal discussion or pairwise ranking, across the top.
    - ✓ Each element is then considered in terms of each criterion.

  - Diagramming
    - ✓ Diagrams, including maps, sketches and transects, summaries data in such a way

that they can be used for different purposes such as planning, field discussion, and analysis and problem identification.
    - ✓ They are useful for opening up discussions between community members and the external team and helping clarify issues and questions.

  - Participatory mapping
    - ✓ Maps can be used to identify the comparative location and importance of different resources within an area.
    - ✓ They can examine a great breadth of subject matter, and allow for a range of different types of map to be produced for one area or for comparative analysis by different groups within the same area.
    - ✓ Maps are also a useful tool to aid communication and to generate the following:
      - Provide a framework for discussion over the relative location of resources
      - Highlight resources of importance, using maps as a spatial guide
      - Raise issues which affect or are affected by these resources
      - Analyze the present status or condition of a location
      - Create a focus for interest in a discussion over resources
      - Social maps can be used to locate houses, services and infrastructure within an area.
      - Maps should be used as a visual stimulant, to identify the parameters faced by local people and to facilitate discussion about the importance people place on infrastructure provision etc.

- The following are **constraints** of PRA
  - The process of carrying out PRA methods requires mutual respect and trust between external facilitators and village participants; this takes time, and requires an understanding of the local culture.
  - Time consuming
  - A common mistake is, projects do not contact local government officials from the start - hence no local government support which is a prerequisite for success.
  - The step from problem identification with participatory PRA methods to follow up project definition is often poorly prepared.
  - Project officers should leave any class and gender biases they may carry at home!
  - The researcher must be able to listen but also to establish dialogue and share own experience.
  - Cultural protocol may be difficult to follow
  - It raises expectations
  - A researcher stays in the village throughout the training session